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The area under the precision-recall curve as a performance metric for rare binary events

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Abstract

1. Species distribution models are used to study biogeographic patterns and guide decision-making. The variable quality of these models makes it critical to assess whether a model's outputs are suitable for the intended use, but commonly used evaluation approaches are inappropriate for many ecological contexts. In particular, unrealistically high performance assessments have been associated with models for rare species and predictions over large geographic extents.
2. We evaluated the area under the precision-recall curve (AUC-PR) as a performance metric for rare binary events, focusing on the assessment of species distribution models. Precision is the probability that a species is present given a predicted presence, while recall (more commonly called sensitivity) is the probability the model predicts presence in locations where the species has been observed. We simulated species at three levels of prevalence, compared AUC-PR and the area under the receiver operating characteristic curve (AUC-ROC) when the geographic extent of predictions was increased and assessed how well each metric reflected a model's utility to guide surveys for new populations.
3. AUC-PR was robust to species rarity and, unlike AUC-ROC, not affected by an increasing geographic extent. The major advantages of AUC-PR arise because it does not incorporate correctly predicted absences and is therefore less prone to exaggerate model performance for unbalanced datasets. AUC-PR and precision were useful indicators of a model's utility for guiding surveys.
4. We show that AUC-PR has important advantages for evaluating models of rare species, and its benefits in the context of unbalanced binary responses will make it applicable for other ecological studies. By not considering the true negative quadrant of the confusion matrix, AUC-PR ameliorates issues that arise when the geographic extent is increased beyond the species' range or when a large number of background points are used when absence information is unavailable. However, no single metric captures all aspects of performance nor provides an absolute index that can be compared across datasets. Our results indicate AUC-PR and precision can provide useful and intuitive metrics for evaluating a model's utility for guiding sampling, and can complement other metrics to help delineate a model's appropriate use.

KEYWORDS
discrimination performance, model assessment, performance metric, receiver operating characteristic curve, species distribution modelling, virtual species

1 | INTRODUCTION

Species distribution models are a primary tool used to map species' habitat suitability and inform conservation and management planning (Guisan et al., 2013). Models can be developed with a wide array of statistical algorithms and vary considerably in the quality and quantity of input data describing species occurrences and environmental conditions. Understanding how methodological decisions and input data affect the credibility of model output requires rigorous criteria for model assessment and performance evaluation (Vaughan & Ormerod, 2005; Fourcade, Besnard, & Secondi, 2018). Nevertheless, the most widely used performance metrics have important limitations for ecological studies (Lobo, Jimenez-Valverde, & Real, 2008; Jiménez-Valverde, 2014; Lawson, Hodgson, Wilson, & Richards, 2014; Somodi, Lepesi, & Botta-Dukát, 2017), and a positive performance assessment based on these metrics does not guarantee effectiveness for guiding specific management and conservation applications (Guillera-Arroita et al., 2015; Sofaer, Jarnevich, & Flather, 2018).

Many of the difficulties associated with performance assessment arise when species occurrences represent rare events – and most species are rare. In these cases, a model that always predicts the common class (i.e. absence) will yield largely correct classifications (Manel, Williams, & Ormerod, 2001). Performance metrics recognize that these correct predictions do not represent model skill, and that a species' prevalence (the ratio of occupied sites to total sites) determines the probability of being correct by chance (Fielding & Bell, 1997; Lawson et al., 2014). For this reason, the True Skill Statistic (TSS), kappa and similar metrics are prevalence dependent (Vach, 2005; Santika, 2011; Somodi et al., 2017), although they still have challenges with assessing predictions of rare events (Doswell, Davies-Jones, & Keller, 1990). These metrics are calculated based on a specific threshold. The ability to summarize model performance across all possible thresholds is often considered an advantage of the most widely used performance metric, the area under the receiver operating characteristic curve (AUC-ROC). Like performance metrics based on a single confusion matrix (Table 1), AUC-ROC measures discrimination performance, defined as the ability to separate ones from zeros. AUC-ROC represents the probability that a randomly selected presence will have a higher continuous prediction than a randomly selected absence across all thresholds (Hanley & Mcneil, 1982).

TABLE 1 Confusion matrix highlighting the attributes of performance used in calculating AUC-ROC and AUC-PR

	Observed present	Observed absent	
Predicted present	TP	FP	Precision = TP/(TP+FP)
Predicted absent	FN	TN	
	Recall = Sensitivity = TP/(TP+FN)	Specificity = TN/(FP+TN)	

AUC-ROC is the area under the curve of the plot of sensitivity (recall) versus 1-specificity across thresholds. Sensitivity is based on all observed presences (left column), while specificity is based on all observed (or inferred) absences (right column). Thus, AUC-ROC incorporates all quadrants of the confusion matrix. AUC-PR is the area under the curve of the plot of precision versus recall (sensitivity) across thresholds. Precision is based on all predicted presences (top row). Thus, AUC-PR does not incorporate the number of true negatives (TN); cells used in calculating AUC-PR are shaded grey and outlined in dashes.

TP, True positive; FP, False positive; FN, False negative; TN, True negative.

Two limitations of AUC-ROC for species distribution modelling have been highlighted in the literature: variation with geographical extent and reliance on knowledge of true absences. As a study area expands beyond the range margins of the focal species, more locations with correct predictions of low suitability are included, inflating AUC-ROC (Luoto, Pöyry, Heikkinen, & Saarinen, 2005; Elith et al., 2006; Lobo et al., 2008; Barve et al., 2011; Acevedo, Jiménez-Valverde, Lobo, & Real, 2012). For example, even a poor model can predict that a desert shrub will not occur in a rainforest, and model assessments at continental or global scales often include many locations with environmental conditions very dissimilar to those under which a given species occurs. This issue is closely linked to species prevalence because as geographical extent increases beyond a species' range margins, the species becomes rarer. True prevalence is difficult to estimate from presence–absence data because of issues including sampling biases and imperfect detection; when absence data are unavailable, true prevalence is simply unknown (Guillera-Arroita et al., 2015; Leroy et al., 2018). While AUC-ROC is mathematically independent of prevalence (Ferro & Stephenson, 2011; Jiménez-Valverde, 2014; Lawson et al., 2014), in practice it is related to prevalence in presence–absence datasets (reviewed in Santika, 2011). Specifically, models of species that are rare in geographic space or specialists in environmental space often yield high values of AUC-ROC, and it is difficult to disentangle inflation of the performance metric from whether models capture the distributions of these species more appropriately (Jimenez-Valverde, Lobo, & Hortal, 2008; Morán-Ordóñez, Lahoz-Monfort, Elith, & Wintle, 2017). In addition, AUC-ROC and other widely used performance metrics are designed to reflect the trade-off between sensitivity (correct predictions of presence) and specificity (correct predictions of absence), and generally weight sensitivity and specificity equally (Peterson, Papes, & Soberon, 2008; Jiménez-Valverde, 2012). However, different intended uses of distribution models can have different costs for different types of errors (i.e. false presences vs. false absences).

Many species distribution models are based on presence records from centralized databases that collate museum data and other observations. In these cases, species' presence information is often contrasted with background information as the basis for model estimation. When absence data are unavailable, both the inferences that can be drawn from species distribution models and the interpretation of performance metrics necessarily change. Recent work has recognized that multiple analytical frameworks for modelling species presence data can be specified such that they are equivalent, including Maxent, Poisson point process models, generalized linear models and resource selection functions (Fithian & Hastie, 2013; Warton & Aarts, 2013; Renner et al., 2015). Background information characterizes the available environment, and as such, can include the same locations as presence points. It is recommended to use a large number of background locations (Warton & Shepherd, 2010; Northrup, Hooten, Anderson, & Wittemyer, 2013), and as a result, the ratio of ones (presence) to zeros (background, which could be true presences or absences) in data used for model estimation and evaluation will be very low, and sample prevalence is highly unlikely to equal the true, unknown, prevalence (Leroy et al., 2018). In line with the effects of a larger geographic extent in the presence–absence context, AUC-ROC generally increases with the extent from which background locations are drawn for presence–background datasets (VanDerWal, Shoo, Graham, & Williams, 2009). In the presence–background context, AUC-ROC can be interpreted as reflecting the model's ability to discriminate presence locations from background, although its maximum is no longer one (Phillips, Anderson, & Schapire, 2006; Smith, 2013). Despite differences – and difficulties (Jiménez-Valverde, 2012) – in interpretation, commonly used metrics are calculated the same way when based on presence–background or presence–absence data, and we expect the number of locations falling in the 'true negatives' quadrant to be disproportionately high compared to the other three quadrants of the confusion matrix (Table 1). Therefore, assessment metrics that include the true negative quadrant can exaggerate performance (Leroy et al., 2018); the number of true negatives can overwhelm the number of false positives and cause the ratio (i.e. specificity) to converge towards one. When observed absences are not available, several authors have advised using performance metrics that do not incorporate specificity (Hirzel, Le Lay, Helfer, Randin, & Guisan, 2006; Li & Guo, 2013).

The area under the precision-recall curve (AUC-PR) is a model performance metric for binary responses that is appropriate for rare events and not dependent on model specificity (Davis & Goadrich, 2006). It has been widely used in the field of information retrieval, for example, in classifying relevant documents in a search (Lewis, 1992; Schapire, Singer, & Singhal, 1998). Precision-recall curves have also been recognized as useful for classification performance assessment for unbalanced binary responses in bioinformatics (Saito & Rehmsmeier, 2015). AUC-PR has rarely been applied to ecological responses, but its tolerance of skewed data (e.g. many absences and few presences) makes it well suited for quantifying distribution model performance for rare species (Johnson, Chawla & Hellmann 2012). Below, we describe AUC-PR (Figure 1) and use simulations to evaluate its utility as a performance assessment metric for species distribution modelling (Figure 2). We simulated species with three levels of prevalence; each of the three simulated species was represented by a continuous probability surface corresponding to its true habitat suitability. We converted this surface to binary presence and absence in a probabilistic manner, sampled from these binary data and used multiple statistical algorithms to fit species distribution models. For each simulation and model, we calculated performance metrics and explored the following questions:

1. How do AUC-PR and AUC-ROC respond to a changing geographic extent?
2. How do AUC-PR and AUC-ROC change when large unsuitable areas in which the model correctly predicts absence (i.e. many true negatives) are included in the set of predictions?
3. Do AUC-PR, AUC-ROC, precision and calibration reflect a model's utility for guiding survey efforts?

1.1 | Description of AUC-PR

Like AUC-ROC, AUC-PR is a threshold-independent metric that calculates the area under a curve, where the curve is defined by a trade-off between different aspects of performance as the threshold applied to the model's predictions varies (Figure 1). Both ROC and PR curves are functions of the confusion matrix (Table 1). A precision-recall curve shows the precision (the probability that a species is present given a predicted presence) as a function of recall (the probability the model predicts presence in locations with observed presence). Precision, P(observed present | predicted present), reverses the conditionality in recall, P(predicted present | observed present). The trade-off between precision and recall can be useful in selecting a threshold as well as for model evaluation (Schapire et al., 1998; Liu, Berry, Dawson, & Pearson, 2005). For example, precision and recall were used to evaluate models of the occurrence of large fires and visualize trade-offs in performance across thresholds (Stavros, Abatzoglou, Larkin, McKenzie, & Steel, 2014). Recall is equivalent to sensitivity and contributes to AUC-ROC and AUC-PR (Table 1) as well as to other frequently used performance metrics including TSS and kappa. Precision is the same as positive predictive power (or value) and is itself a useful metric, representing the probability a presence is observed given that it was predicted (Fielding & Bell, 1997). Note that this definition of precision differs from the more general statistical definition representing a measure of variability. The *F*-measure (also called the F1 score) is the harmonic mean of precision and recall at a given threshold, and this measure and modifications thereof have been evaluated for assessing performance and selecting thresholds for species distribution models (Wenkai & Qinghua, 2013; Liu, Newell, & White, 2016; Leroy et al., 2018).

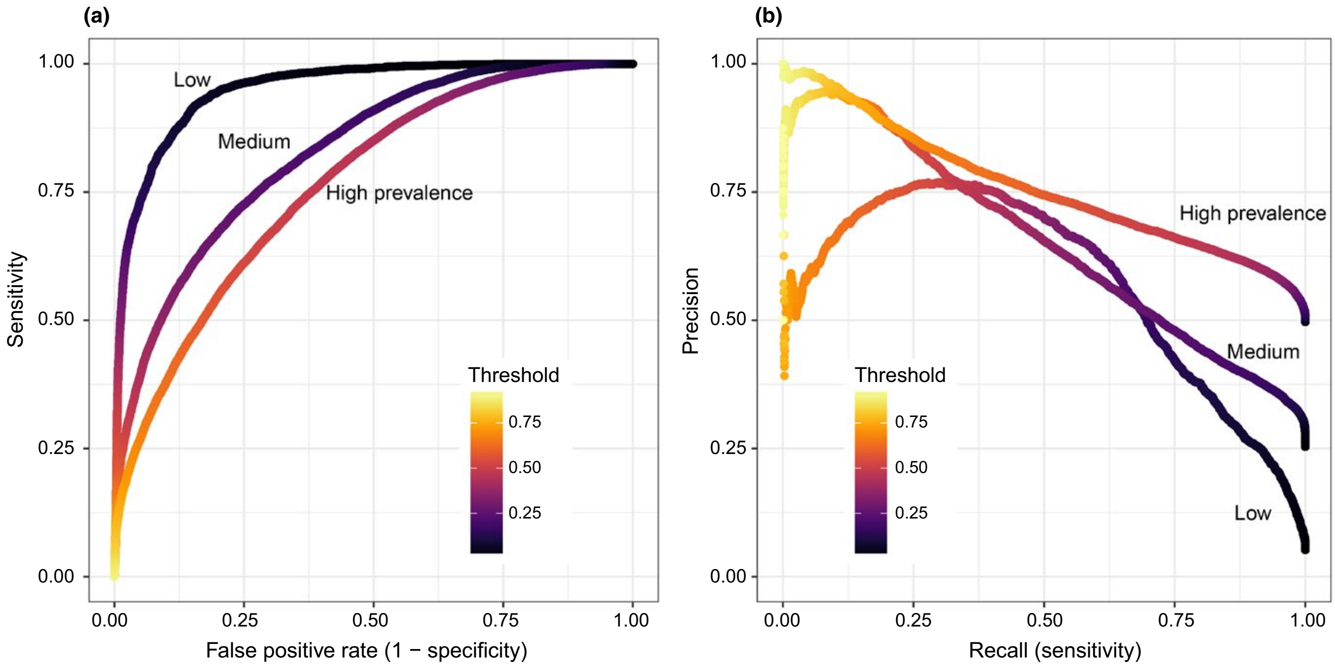

FIGURE 1 Comparison of Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves. (a) ROC curves for ridge regression models from the first replicate simulation for each prevalence level. AUC-ROC is maximized in the upper left corner. (b) PR curves for the same ridge regression models and data. AUC-PR is maximized in the upper right corner. PR curves are more variable in shape than ROC curves

AUC-PR is calculated as the area under the precision-recall curve, where each point on the curve is defined by a different value of the threshold to convert continuous to binary predictions. AUC-PR does not depend on the number of true absence observations (Table 1). While AUC-ROC is maximized by a curve in the upper left-hand corner, AUC-PR is maximized in the upper right-hand corner, reflecting the norm of placing sensitivity on the *y*-axis in a receiver operating characteristic curve but on the *x*-axis in a precision-recall curve (Figure 1a,b). AUC-PR has a more variable shape than AUC-ROC and is not guaranteed to form a monotonic curve.

AUC-PR varies on a scale from zero to one, with random performance equal to sample prevalence in the focal dataset. There is a minimum value of AUC-PR that increases with prevalence because some region of the precision-recall space becomes unattainable even by the worst model (Boyd, Costa, Davis, & Page, 2012). For example, if prevalence is 0.5 and a (bad) model always predicts a 1, recall (sensitivity; the probability of predicting a 1 given an observation of 1) will be 1. Precision, the probability that 1 is observed given 1 is predicted, will be 0.5 (whereas specificity would be 0); this non-zero precision value (i.e. 0.5 in this example) creates a minimum AUC-PR. Minimum AUC-PR with a prevalence of 0.5 is 0.31, following equations in Boyd et al. (2012). Adjusting AUC-PR for its minimum value may make the performance metric more comparable across datasets that differ in prevalence.

2 | MATERIALS AND METHODS

Figure 2 summarizes the materials and methods section.

2.1 | Simulated species

For each of three prevalence levels (low, medium and high), we simulated a continuous probability surface across the conterminous United States (U.S.), reflecting a species' true habitat suitability (Figure 3). The continuous suitability depended on three bioclimatic variables: mean annual temperature (bio1), maximum temperature of the warmest month (bio5) and annual precipitation (bio12). Bioclimatic variables were obtained from WorldClim v1.4 at a 5-min spatial resolution (Hijmans, Cameron, Parra, Jones, & Jarvis, 2005). We used the virtualspecies package (Leroy, Meynard, Bellard, & Courchamp, 2016) in R (R Core Team 2017) to generate a Gaussian-shaped response to each variable and multiplicatively combine each response function to generate the true continuous suitability surface. For all three prevalence levels, suitability peaked at the same values (bio1 = 20°C; bio5 = 45°C; bio12 = 150 mm), making suitability highest in the Southwestern U.S. (Figure 3, top row). The standard deviation of each Gaussian-shaped response curve was increased to yield increasing species prevalence (bio1 = 8, 20, 46 °C; bio5 = 8, 20, 46 °C; bio12 = 77, 173, 460 mm for simulated low, medium and high species prevalence respectively; Supporting information Supporting Information Figure S1). These parameter values were selected to yield simulated species with prevalence in the Western U.S. (defined here as west of the 100th meridian) of approximately 0.05 (low), 0.25 (medium) and 0.5 (high) after the conversion of continuous values to binary presences and absences.

To generate a presence–absence value for each grid cell in the study area, we drew a Bernoulli sample using the probability of presence from

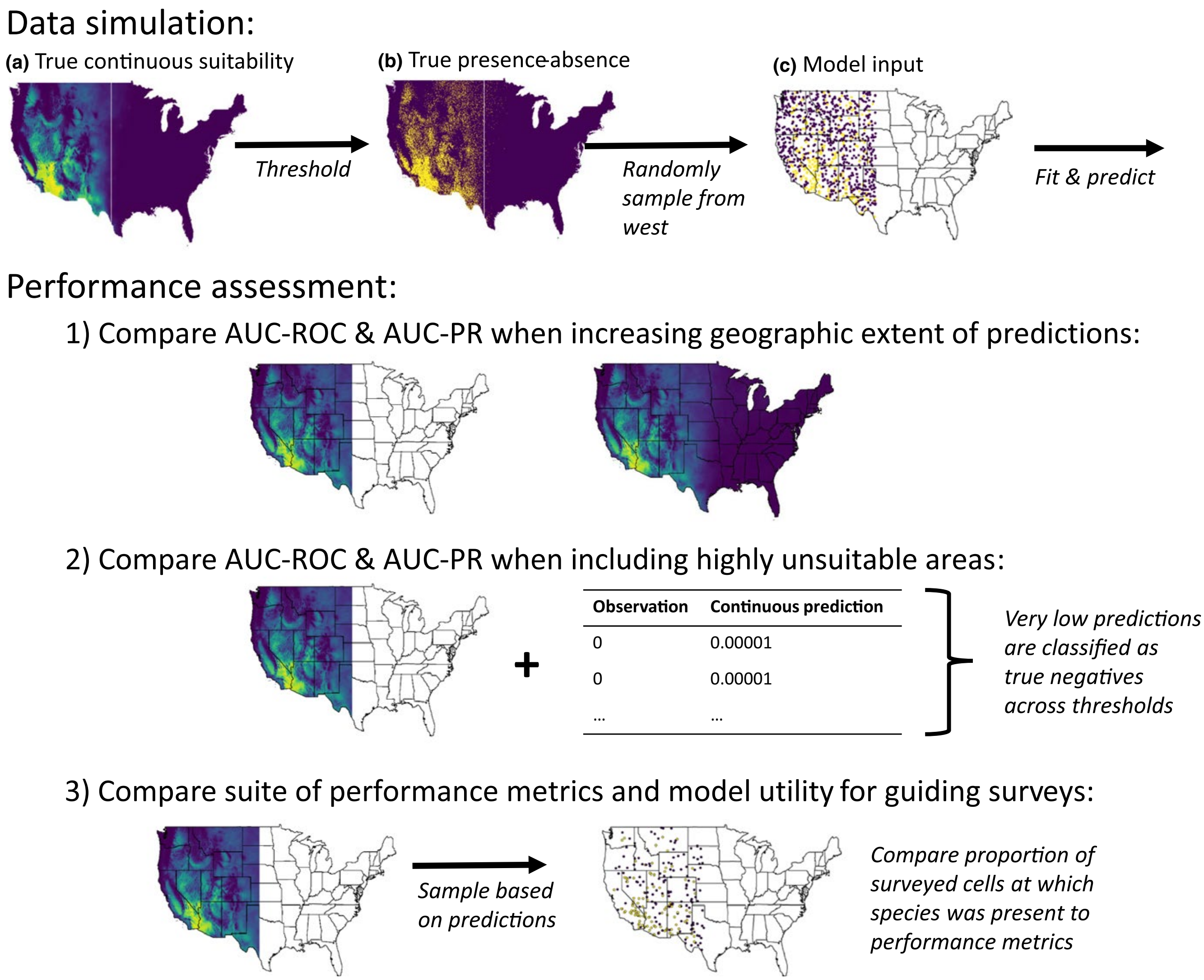

FIGURE 2 Overview of simulation workflow and questions addressed. Top row: (a) For each level of species prevalence, a true continuous suitability surface was simulated based on three climate variables. (b) The continuous suitability was converted into presence–absence information by drawing from a Bernoulli distribution. (c) A random sample within the Western U.S. (n = 1,000 grid cells) was then drawn from the presence–absence layer for model estimation; these locations used in model fitting were excluded from all calculations of performance metrics. (1) We compared AUC-ROC and AUC-PR based on predictions to the Western U.S. versus the conterminous U.S. to evaluate the effects of increasing geographic extent (Figure 4a). (2) We evaluated the effects of including large, highly unsuitable geographic areas in the prediction extent by adding rows to the predictions with very low predicted suitability and observed absence. These rows were classified as true negatives across the range of thresholds in calculations of AUC-ROC and AUC-PR. Enough rows were added to double the size of the dataset (Figure 4b). (3) We used each model as the basis for two types of survey designs: one that selected the highest ranked sites, and one that weighted the probability of sampling a grid cell according to its continuous predicted suitability. Model utility for survey design was defined as the proportion of sampled locations in which the simulated species was present; this proportion was compared to each performance metric to evaluate how well it represented model utility for search (Figure 5)

the simulated continuous surface. The resulting 0/1 observations were treated as 'true' presence–absence information in subsequent analyses (Figure 3, bottom row). This binary presence–absence layer was sampled to provide data for model estimation. We used the randompoints function within the dismo R package (Hijmans, Phillips, Leathwick, & Elith, 2017) to draw 1,000 samples from cells in the Western U.S. Sampling was restricted to the Western U.S. because our simulated species were primarily Western, and we were interested in comparing how AUC-ROC and AUC-PR responded when generating predictions across the Western versus the conterminous U.S. This process was repeated three times for each species-prevalence level, such that the same nine sets of input data (3 species × 3 replicates each) were used for fitting models with different algorithms. The number of presences in the 1,000 presence–absence estimation samples were approximately 55, 260 and 515, from low to high prevalence respectively.

We also evaluated the behaviour of AUC-PR with presence–background data, focusing on our first question regarding the effect of an increasing geographical extent and our third question regarding

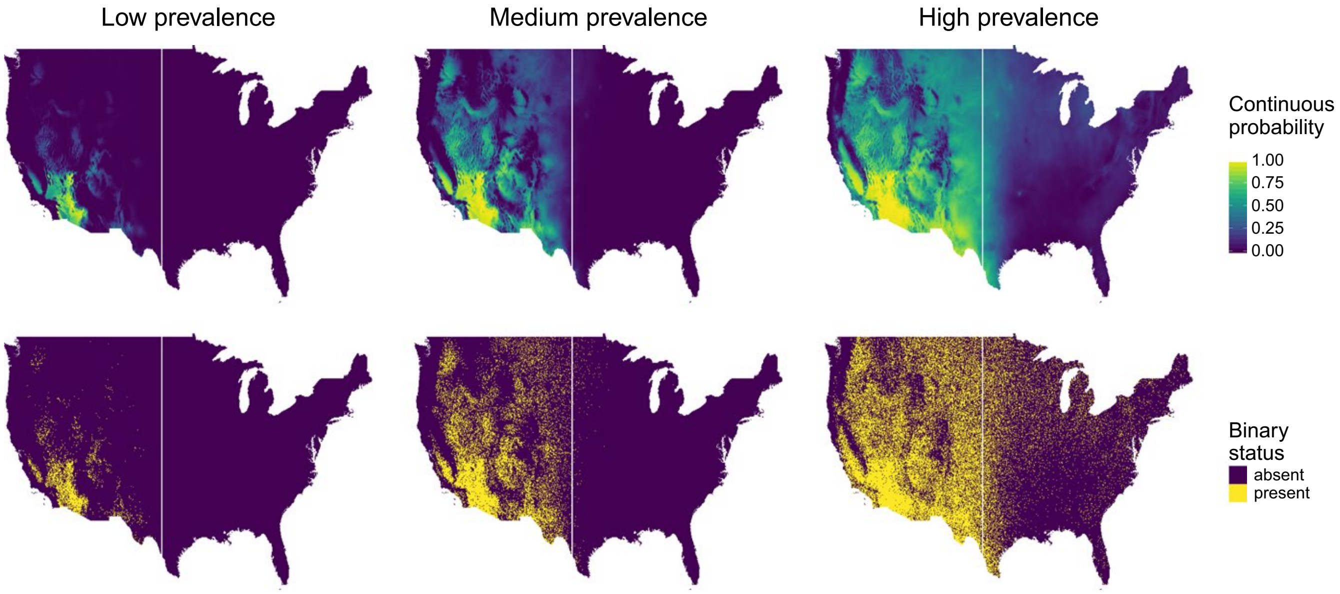

FIGURE 3 Simulated species, generated to exhibit different levels of prevalence, from relatively rare (left column) to common (right column). Continuous probabilities (top row) were based on three climate variables and were probabilistically converted to presence–absence data (bottom row). The vertical white line at the 100th meridian separates the Western and Eastern U.S.

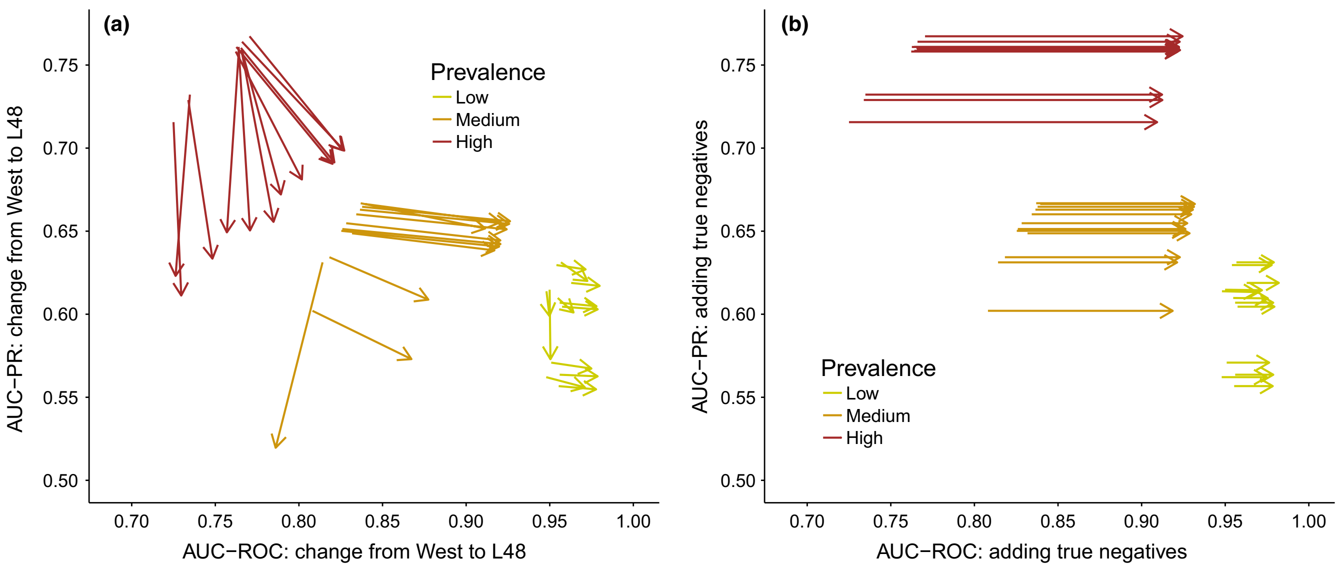

FIGURE 4 (a) When increasing the geographical extent from the Western U.S. to the conterminous U.S. (i.e. lower 48 states; L48), AUC-PR declined while AUC-ROC generally increased. Arrow tails represent values of AUC-PR and AUC-ROC calculated over the Western U.S., while arrow heads represent metric values calculated over the L48. At each prevalence level, there is one arrow for each of the four algorithms based on each of the three replicated simulations (i.e. 12 arrows per prevalence level). (b) The addition of a large number of true negatives, representing predictions made in highly unsuitable areas, caused a substantial increase in AUC-ROC but did not alter AUC-PR. Arrow tails correspond to metric values calculated over the Western U.S., while arrow heads correspond to metric values calculated after doubling the size of the test dataset by adding rows with very low predicted continuous suitability and observed absence; these became true negatives across the gradient of threshold values

assessing utility for guiding surveys. We sampled from our low prevalence virtual species based on occurrence locations of red brome *Bromus rubens*, and used a target-group background approach (Phillips et al., 2009) based on occurrence records of grass species exotic to the U.S. (see Supporting Information Figure S2) to mimic sampling biases encountered in presence–background datasets.

2.2 | Species distribution models

We used four algorithms to fit species distribution models: two machine learning methods (random forest and boosted regression trees) and two types of penalized generalized linear models (lasso and ridge regression). These methods have been shown to perform well for species distribution

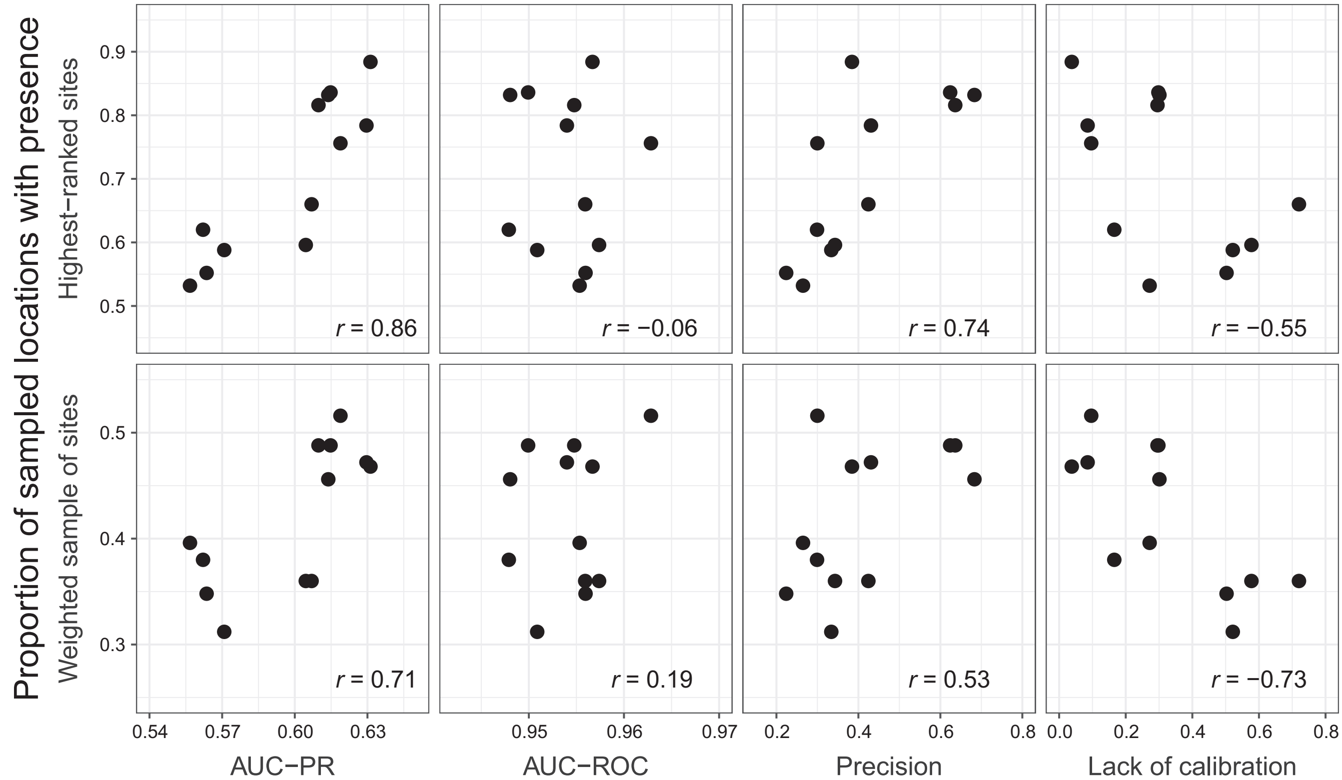

FIGURE 5 AUC-PR, precision and model calibration provided useful assessments of model performance for guiding surveys for new populations. AUC-PR was the metric that best reflected performance when surveys were directed towards the locations with the highest predicted continuous suitability (top row). Precision was also useful in this context. For surveys based on a design that weighted sites according to the predicted continuous suitability (bottom row), model calibration was most closely related to survey performance, with AUC-PR also showing utility. AUC-ROC did not provide a useful assessment of model performance for guiding either type of survey effort

modelling and are relatively (or very) robust to correlated predictor variables (Elith et al., 2006; Dormann et al., 2013). In the randomForest R package (Liaw & Wiener, 2002), we increased the number of trees to 1,000 but otherwise used the default settings. Boosted regression trees were implemented via the dismo package (Hijmans et al., 2017) with a learning rate of 0.001, a tree complexity of 3 and a bag fraction of 0.5. Lasso and ridge logistic regression models were fit via the glmnet package (Friedman, Hastie, & Tibshirani, 2010). For simplicity, we did not include squared terms or interactions among variables in lasso or ridge regression models. Cross-validation was used to select the magnitude of the penalty parameter in lasso and ridge regression models, and predictions were based on the highest penalty that produced an error rate within one standard error of the minimum error rate. Each statistical algorithm was fit based on all 19 bioclimatic variables because the true drivers of habitat suitability are not known in practice. For each algorithm, species and replicate simulation, we used the maximum of sensitivity plus specificity criterion to identify an optimal threshold based on the estimation data; this was implemented via the SDMTools (VanDerWal, Falconi, Januchowski, Shoo, & Storlie, 2014) R package. We applied this threshold to continuous predictions to derive predicted presence and absence for computation of model precision (Table 1). Precision is based on a single fixed threshold and represents the probability of observing presence given that presence was predicted; it was, therefore, expected to be useful for assessing model utility for informing survey designs.

We compared performance metrics to investigate (Figure 2): (1) the effect of increasing the geographical extent on AUC-PR and AUC-ROC, (2) the effect of including highly unsuitable areas on AUC-PR and AUC-ROC and (3) the utility of AUC-PR, AUC-ROC, precision and a calibration metric for assessing model performance for guiding future surveys. Performance metrics were calculated using the PRROC (Grau, Grosse, & Keilwagen, 2015), SDMTools (VanDerWal et al., 2014) and MODEvA (Barbosa, Brown, Jimenez-Valverde, & Real, 2016) R packages. We also visualized how the magnitude of minimum adjustment for AUC-PR varied across prevalence levels in our simulations, plotting AUC-PR and minimum-adjusted AUC-PR (Supporting Information Figure S3).

2.3 | Question 1: Effect of increasing geographical extent

We predicted that AUC-PR would be more robust than AUC-ROC to the effects of increasing geographical extent because AUC-PR does not incorporate true negatives (Table 1). We compared the change in AUC-ROC and AUC-PR when predicting species distributions for the Western U.S. versus across the conterminous U.S. Models were estimated based on data from 1,000 randomly selected sites in the Western U.S., and AUC-ROC and AUC–PR were computed on all other grid cells (either Western U.S. only or conterminous U.S.), omitting the 1,000 cells used for model fitting. Therefore, the training extent remained constant while the test extent differed.

We evaluated the sensitivity of our results to extrapolation in environmental space. Multivariate Environmental Similarity Surfaces (MESS; Elith, Kearney, & Phillips, 2010) were calculated for each species and replicate simulation, and we compared the change in AUC-ROC and AUC-PR when limiting the increase in geographical extent to grid cells in the conterminous U.S. with non-negative MESS values (i.e. excluding areas of extrapolation). We also evaluated the sensitivity of our findings to the use of presence–background data for model estimation (see Supporting Information).

2.4 | Question 2: Effect of including highly unsuitable areas

To isolate the effects of including highly unsuitable areas in the study extent (and hence decreasing prevalence), we augmented the Western U.S. dataset used for model testing with additional data with predicted suitability = 0.00001 and observed absence. These very low predicted suitabilities become predicted absences across the gradient of thresholds considered in the calculations of AUC-ROC and AUC-PR, thereby adding true negatives to the confusion matrix. This addition of true negatives corresponded to a situation in which the geographic extent for model prediction included many locations with conditions very divergent to those within a species' range, such that models could correctly predict low suitability. Although the addition of these observations was artificial, in that it did not correspond to real geographic locations in which the simulated species was present or absent, this approach was designed to isolate the effects of adding true negatives. In contrast, increasing the geographic extent could add to all quadrants of the confusion matrix, particularly for the simulated species with higher prevalence (Supporting Information Figure S4). We added enough of these observations to double the size of the dataset (n = 57,465 observations), approximating the increase (n = 60,321 grid cells) from the Western to conterminous U.S. at the focal spatial resolution.

2.5 | Question 3: Assessing model performance for guiding surveys

An important use of species distribution models is for guiding subsequent survey efforts, particularly for rare (Guisan et al., 2006) and invasive (Crall et al., 2013) species. We evaluated how AUC-PR, AUC-ROC, precision and calibration performance related to a model's ability to effectively guide search, focusing on the low prevalence simulated species within the Western U.S. We limited this analysis to a single species and extent to avoid confounding our findings with changes in performance arising from changes in species prevalence, and to represent a comparison among model algorithms when one method is to be selected to guide a survey design.

For each model, we sampled 250 locations based on two different approaches; both approaches excluded locations used in model estimation (i.e. where presence–absence status was already known). The first approach selected the grid cells with the highest predicted continuous suitability. In the second approach, we applied an unequal probability sampling design using the predicted continuous suitability as the sampling weight. The first method selected the highest ranked sites, and we predicted that discrimination metrics (which capture ranking performance) would be useful for predicting survey performance when targeting the highest ranked locations. The second method relied on a proportional relationship between continuous model predictions and true suitability, a criterion intermediate between ranking ability and perfect calibration (Guillera-Arroita et al., 2015). We predicted that a calibration metric would best reflect each model's ability to effectively guide search based on a weighted sampling design. As a measure of (lack of) calibration, we calculated the absolute difference between the slope of Miller's calibration plot and one, as a perfectly calibrated model would have a slope of one (Miller, Hui, & Tierney, 1991; Pearce & Ferrier, 2000). For each of the three binary presence–absence layers of the lowest prevalence species (arising from the three replicate runs) and for each model algorithm, we calculated the proportion of these new survey locations with true presences. We used this proportion to represent each model's utility for search. We plotted this proportion against each performance metric and calculated Pearson's correlation coefficient for each relationship. We also assessed the sensitivity of our finding to the use of presence–background data for model estimation, applying both survey approaches to predictions from those models (see Supporting Information).

3 | RESULTS

3.1 | Question 1: Effect of increasing geographical extent

AUC-ROC and AUC-PR showed divergent responses to an increase in the geographic extent of model predictions (Figure 4a). We found increases in AUC-ROC with an increasing geographic extent, in line with results from previous work (Elith et al., 2006; Jimenez-Valverde et al., 2008; Lobo et al., 2008). In contrast, increasing the geographic extent from the Western U.S. to the conterminous U.S. caused AUC-PR to decline, with a greater decline seen in the simulated species with higher prevalence (Figure 4a). This result arose because as prevalence increased, more cells in the Eastern U.S. were suitable for the species. Therefore, across the full range of possible thresholds, sets of binary predictions and observations fell within all cells of the confusion matrix (Figure S4). The decline in AUC-PR with increasing extent was greatest for the random forest algorithm, which had more false-positive predictions at low threshold values compared with other algorithms.

Our results emphasize the advantages of AUC-PR over AUC-ROC for unbalanced datasets because the primary difference in the distribution of continuous predicted suitability was an increase in very low values when predicting to the conterminous U.S. (Supporting Information Figure S5). In other words, models appropriately predicted low suitability outside the simulated species' ranges. Increasing the imbalance (i.e. the number of zeros relative to ones) in the validation dataset resulted in a higher ROC curve, while the precision-recall curve was effectively unchanged (Supporting Information Figure S5).

Our findings were similar when excluding grid cells with extrapolation in environmental space as measured by MESS (Supporting Information Figure S6); AUC-PR never increased, while AUC-ROC increased for the low and medium prevalence species, but generally declined for the high prevalence species. Similarly, for models based on presence–background data, AUC-PR never increased with an increase in geographical extent, while AUC-ROC increased for three of the four model algorithms (Supporting Information Figure S7).

3.2 | Question 2: Effect of including highly unsuitable areas

AUC-PR does not incorporate true negatives into its calculations, and as expected, was not affected by the doubling of the size of the predicted dataset to include locations with very low predicted suitability and observed absence (Figure 4b). In contrast, AUC-ROC increased substantially under this scenario, in some cases enough to change a qualitative assessment of model quality. AUC-ROC showed a larger increase for the more prevalent simulated species. Adding only true negatives highlighted the advantages of AUC-PR compared to AUC-ROC for imbalanced datasets; these advantages are particularly relevant for multispecies studies in which the geographic extent is not tailored to each species' range.

3.3 | Question 3: Assessing model performance for guiding surveys

AUC-PR provided a useful assessment of a model's utility for guiding search. This was particularly true when survey efforts targeted the sites with the highest predicted continuous suitability (Figure 5, top row). Such a survey design relies on a model's ability to rank sites, and as such, discrimination metrics such as AUC-PR and AUC-ROC were expected to be useful. While AUC-PR was closely correlated with the proportion of top-ranked sites at which the simulated species was present ($r = 0.86$), AUC-ROC was not related to this measure of survey success ($r = -0.06$). Precision was also positively correlated with survey success based on top-ranked sites ($r = 0.74$). For the survey design that sampled sites by weighting by the continuous predicted suitability (Figure 5, bottom row), well-calibrated models could most effectively guide sampling ($r = -0.73$; 'lack of calibration' is the difference between a model's calibration plot slope and 1). This is in line with the expectation that the degree to which model predictions are proportional to true suitability will shape the success rate of the weighted sampling design.

For models based on presence–background data, relationships between performance metrics and utility for guiding surveys were weaker (Supporting Information Figure S8; $r \leq 0.4$ for AUC-PR, AUC-ROC and precision). Model calibration was the best predictor of survey success ($r = -0.56$ for selection of top-ranked sites and $r = -0.79$ for weighted sample; as before, correlation is with 'lack of calibration'). Our calculations of calibration relied on knowledge of true simulated presence absence data, and while methods exist to evaluate calibration of presence–background data (e.g. Phillips & Elith, 2010), challenges in assessment and interpretation remain.

4 | DISCUSSION

The area under the precision-recall curve (AUC-PR) provides a relevant performance metric for models of rare binary events. Compared with AUC-ROC, AUC-PR is robust to unbalanced datasets (Supporting Information Figure S5). Our simulations highlight specific advantages of AUC-PR for assessing species distribution models: first, AUC-PR does not increase with an increasing geographical extent or the inclusion of many highly unsuitable locations within the study area (Figure 4), and second, when based on presence–absence information, AUC-PR reflects a model's ability to guide surveys for new populations (Figure 5). These features arise because AUC-PR does not incorporate correct predictions of absence (Table 1). Collectively, the advantages of AUC-PR address major limitations of AUC-ROC for ecological studies of rare events (Lobo et al., 2008), although as with other performance metrics, its use and interpretation in the context of presence–background data is more challenging than when absence information is available. We found that AUC-PR retained some advantages when presence locations were compared with background samples, rather than absence data; in particular, it never increased with an increasing geographic extent. AUC-PR will be a useful performance assessment metric for many applications of species distribution models as well as for other ecological models with binary responses of rare events. Species distribution models are highly variable in the quality of their outputs, and a given model may be useful for some purposes but not others. By evaluating models with performance metrics that reflect utility for the intended application, modellers can provide managers and practitioners with more credible and relevant summaries of a model's ability to inform decision-making.

Our simulations varied species prevalence and compared how assessments of model performance based on AUC-ROC and AUC-PR were affected. Prevalence varied among simulated species (low, medium and high) and declined for all species when enlarging the geographic extent to encompass the entire conterminous U.S. Our simulations found that AUC-ROC was higher for rarer species, in line with previous findings (Luoto et al., 2005; Elith et al., 2006; Evangelista et al., 2008; Jimenez-Valverde et al., 2008), and also generally confirmed increases in AUC-ROC with an increasing geographic extent (Lobo et al., 2008; Acevedo et al., 2012). We note that these changes arise from the manner in which AUC-ROC varies among datasets with different levels of prevalence and are confounded with any biologically driven differences in model performance. Our simulations allowed model sensitivity and specificity to change in a realistic manner as the datasets became more unbalanced (i.e. the species became rarer). In contrast, simulations that vary prevalence while holding sensitivity and specificity constant show no change in receiver operator curves, but nevertheless also highlight the advantages of precision-recall curves for evaluating models of rare events (Saito & Rehmsmeier, 2015).

No single model performance metric captures all potential dimensions of model quality, and a key challenge is linking the methods used to guide model validation and assessment to a model's intended use. Our work highlights the applicability of AUC-PR and precision for models based on presence–absence data that are designed to guide surveys for new populations. We found that the type of survey design affected the relevant performance metric; AUC-PR was most relevant for assessing a model's ability to identify the most suitable sites, while calibration was a better predictor of performance for a weighted survey design based on a model's continuous probabilities. For assessing performance in identifying the most suitable sites, AUC-PR and precision are complementary, with AUC-PR providing a threshold-independent metric and precision providing a threshold-specific assessment. Precision is particularly intuitive, and the probability that a location is occupied given that the model predicts presence is clearly relevant for designing new surveys. However, an important consideration for the use of models to guide search and surveillance is whether realized or potential distributions are most applicable. For example, mapping the current extent of an invasive species can help locate and control previously unknown populations, while mapping potential distributions is more relevant for developing watch lists and other risk assessment tools. Our simulations were designed to represent search within a species' currently occupied range and demonstrated the utility of AUC-PR and precision. However, both AUC-PR and precision penalize false positives, and so may be less appropriate for modelling species' potential ranges. This is because locations may be correctly identified as having high potential suitability, despite being currently unoccupied. AUC-ROC has similarly been recognized to be most useful for modelling realized rather than potential ranges (Jiménez-Valverde, 2012). This nuance highlights the need to consider how different types of errors are incorporated in each performance metric, and which types of error are most critical for a given intended use.

Ideally, a performance metric would provide researchers and practitioners with a universal and objective criterion to determine whether a given model will be useful for the target application. However, there is often a disconnect between the technically correct and the frequently applied interpretation of performance metric values. In particular, rules of thumb have been suggested for AUC-ROC (Swets, 1988). For presence–absence data, these rules of thumb have been applied to compare models with shared estimation and evaluation data (i.e. across algorithms for a single species) as well as among models based on different presence–absence data (e.g. across species with rigorous survey data). Similarly, performance metrics are often compared for models based on presence–background information, both within and among datasets (e.g. across species with information from occurrence databases). AUC-ROC has been criticized even for the best-case scenario: comparison among algorithms applied to a single presence–absence dataset (Hand, 2010), a criticism that also applies to AUC-PR (Parker, 2011). This is because in practice, differences in the distribution of model predictions lead to different assumed relative costs of false-negative versus false-positive errors (Hand, 2010; Jiménez-Valverde, 2014). An important avenue for additional research is to evaluate how the degree of divergence in the distribution of model predictions relates to the differences in assumed costs, and hence to the degree to which performance metrics are not directly comparable. The distribution of model predictions will depend on prevalence as well as on the distribution of predictor variables, and as such, comparisons of metric values among datasets will be more problematic than for those within datasets (Jiménez-Valverde, Acevedo, Barbosa, Lobo, & Real, 2013; Jiménez-Valverde, 2014).

All performance metric values are contingent on the dataset with which they were computed, and comparison of the value of performance metrics across datasets that diverge in attributes – especially species prevalence – is generally subjective (Yackulic et al., 2012; Lawson et al., 2014). Comparisons of metric values across datasets are particularly problematic for models fit to presence–background data, an issue that is well recognized for AUC-ROC (Lobo et al., 2008; Elith et al., 2011; Smith, 2013). Our simulations highlighted similar challenges for AUC-PR, which was a good predictor of a model's ability to identify the most suitable locations when models were based on presence–absence data (Figure 5), but not when models were based on presence–background data (Supporting Information Figure S8). For presence–background data, AUC-PR should have an advantage over AUC-ROC in that it was not inflated by a large number of values in the (inferred) true absence quadrant of the confusion matrix in model evaluation datasets and did not increase with an increase in geographical extent (Supporting Information Figure S7). Nevertheless, AUC-PR should not be considered a panacea for the lack of observed absences.

In our simulations, AUC-PR generally increased with simulated species prevalence, while AUC-ROC was highest for the rarest simulated species and decreased with prevalence. For AUC-PR, this pattern is expected given that there is a minimum value that increases with prevalence. Adjusting for the minimum value has been suggested because it can make comparisons across datasets and species more appropriate (Boyd et al., 2012). Additional comparisons of AUC-PR and minimum-adjusted AUC-PR are needed. Depending on a study's objectives, either unadjusted or minimum-adjusted AUC-PR may be most appropriate; when prevalence is low, there will be little difference (Supporting Information Figure S3).

Species distribution models can play an important role in informing spatially explicit conservation and management planning, but there is often a gap between model production and uptake into decision-making (Guisan et al., 2013). Tailoring model performance assessments towards a given application of interest, and clearly communicating the results, should help managers and practitioners determine when and how a model will be useful. We show that AUC-PR provides a valuable assessment metric for comparison among models when data are imbalanced (Table 1; Supporting Information Figure S5). This makes it a useful metric for rare species and for multispecies studies in which the geographic extent is larger than individual species' ranges. When presence–absence data are available, AUC-PR provides a useful comparison of models' ability to guide surveys to the most suitable sites for a given species; we found that assessing model calibration was more relevant for weighted survey designs. AUC-PR may also lessen – but not eliminate – issues in comparisons among datasets that lack observations of absence because these datasets often yield a large proportion of true negatives in the confusion matrix, which are of uncertain value and not incorporated into AUC-PR. AUC-PR does incorporate false positives and so is not independent of knowledge of absence, nor will it address systematic biases arising from imperfect detection. We suggest that AUC-PR can complement other model evaluation and performance assessment approaches, and that the advantages of AUC-PR for unbalanced datasets will make it broadly relevant for binary ecological responses.

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AUTHORS' CONTRIBUTIONS

H.R.S., J.A.H. and C.S.J. designed the study. H.R.S. conducted analyses and wrote draft manuscript which all authors edited.

DATA ACCESSIBILITY

Code to simulate and analyse data is publicly available at https://github.com/hsofaer/AUC-PR, https://doi.org/10.5281/zenodo.2228178 (hsofaer, 2018).

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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